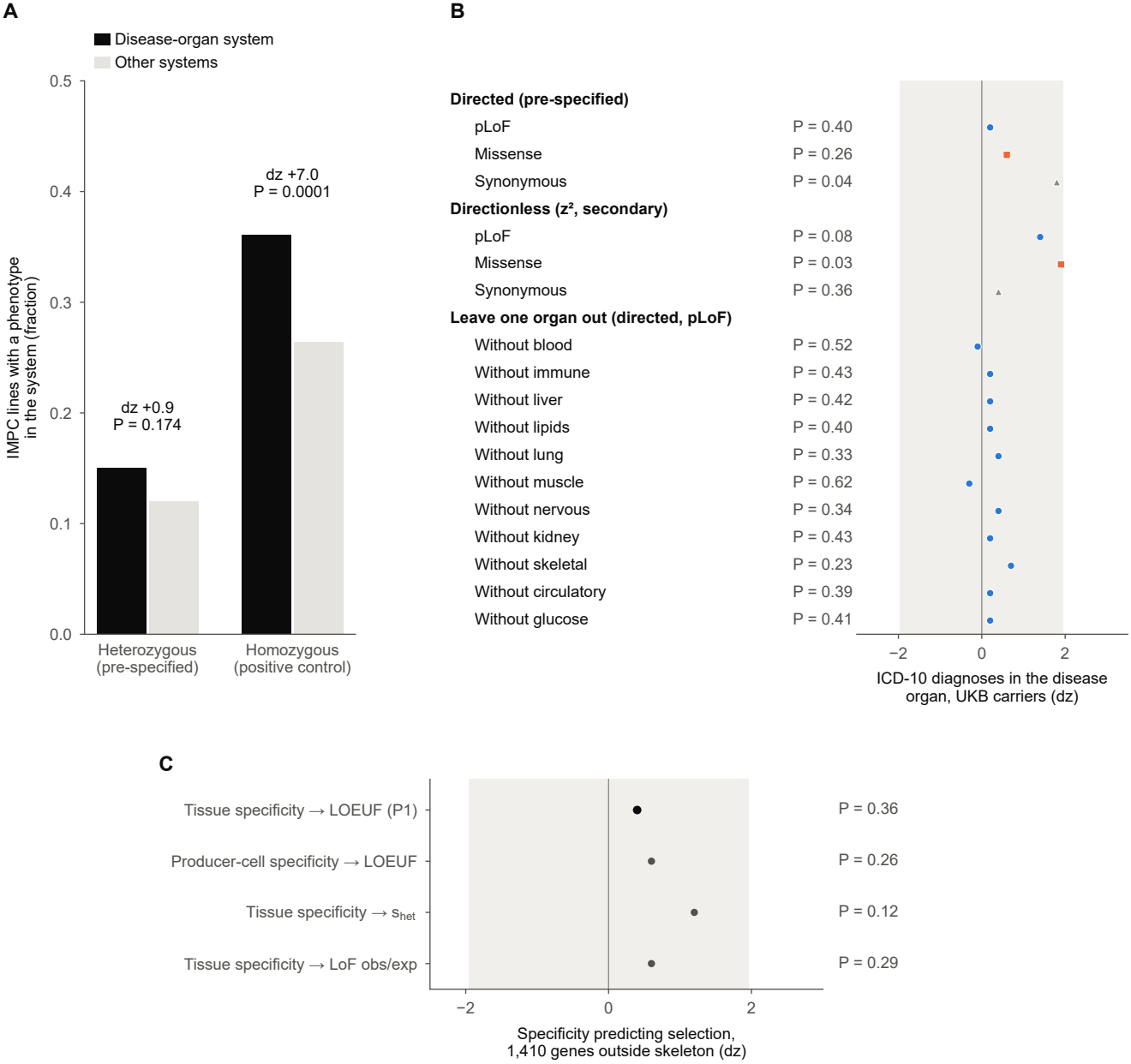
A
Disease-organ system
Other systems
IMPC lines with a phenotype in the system (fraction)
0.5
0.4
0.3
0.2
0.1
0.0
dz +7.0
P = 0.0001
dz +0.9
P = 0.174
Heterozygous (pre-specified)
Homozygous (positive control)
B
Directed (pre-specified)
pLoF
P = 0.40
Missense
P = 0.26
Synonymous
P = 0.04
Directionless (z², secondary)
pLoF
P = 0.08
Missense
P = 0.03
Synonymous
P = 0.36
Leave one organ out (directed, pLoF)
Without blood
P = 0.52
Without immune
P = 0.43
Without liver
P = 0.42
Without lipids
P = 0.40
Without lung
P = 0.33
Without muscle
P = 0.62
Without nervous
P = 0.34
Without kidney
P = 0.43
Without skeletal
P = 0.23
Without circulatory
P = 0.39
Without glucose
P = 0.41
−2
0
2
ICD-10 diagnoses in the disease organ, UKB carriers (dz)
C
Tissue specificity → LOEUF (P1)
Producer-cell specificity → LOEUF
Tissue specificity → s_het
Tissue specificity → LoF obs/exp
P = 0.36
P = 0.26
P = 0.12
P = 0.29
−2
0
2
Specificity predicting selection, 1,410 genes outside skeleton (dz)